MATH2241 Introduction to Mathematical Analysis
Review Session 1: Practice Problems

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Problem 1. Let A be a non-empty subset of $\mathbb{R}$ and let α be an upper bound of A. Suppose that for every $n \in \mathbb{N}$, there exists $a_n \in A$ such that $\alpha - \frac{1}{n} < a_n$. Prove that $\alpha = \sup A$.

Problem 2. Give an example of each of the following, or prove that such a situation is impossible:

(a) sequences $\{a_n\}$ and $\{b_n\}$, where $\{a_n\}$ converges, $\{b_n\}$ is unbounded, and $\{a_n b_n\}$ converges.

(b) sequences $\{a_n\}$ and $\{b_n\}$, where $\{a_n\}$ is unbounded, $\{b_n\}$ converges, and $\{a_n - b_n\}$ is bounded.

(c) sequences $\{a_n\}$ and $\{b_n\}$, where $\{a_n\}$ converges, $\{b_n\}$ diverges, and $\{a_n + b_n\}$ converges.

(d) sequences $\{a_n\}$ and $\{b_n\}$, which both diverge, but $\{a_n + b_n\}$, $\{a_n b_n\}$, and $\{a_n / b_n\}$ all converge.

Problem 3. Let $a > 1$. Define a sequence $\{x_n\}$ by

$$x_1 = a, \qquad x_{n+1} = \frac{1}{2}\left(x_n + \frac{a}{x_n}\right) \quad \text{for any } n \in \mathbb{N}.$$

(a) Prove by induction that $x_n > \sqrt{a}$ for any $n \in \mathbb{N}$.

(b) Prove that $\{x_n\}$ is decreasing. Hence show that $\{x_n\}$ converges. Find also the limit.

Problem 4. Let $x_1 := a > 0$ and $x_{n+1} := x_n + 1/x_n$ for $n \in \mathbb{N}$. Determine whether (x_n) converges or diverges.

Problem 5. Let $0 < \alpha < 1$ and let $\{a_n\}$ be a sequence which satisfies

$$|a_{n+2} - a_{n+1}| \leq \alpha |a_{n+1} - a_n| \qquad \text{for all } n \in \mathbb{N}.$$

Prove that $\{a_n\}$ is a Cauchy sequence (and hence converges).

Problem 6. Determine whether each of the following series converges. Justify your answers.

(a) $\sum \dfrac{n^2}{n!}$

(b) $\sum \left[\sqrt{n+1} - \sqrt{n}\right]$

(c) $\sum \dfrac{n!}{n^n}$

(d) $\displaystyle\sum_{n=2}^{\infty} \frac{1}{\sqrt{n}\log n}$

Problem 7.

(a) Suppose the series $(s_n = \sum_{k=1}^{n} a_k)$ is absolutely convergent. Prove that the series $\left(t_n = \sum_{k=1}^{n} a_k^3\right)$ is convergent.

(b) More generally, what can be said about $\sum a_n^k$ for other integers k?

Problem 8.

(a) Find a series $\sum a_n$ which diverges by the Root Test but for which the Ratio Test gives no information.

(b) Find a series $\sum a_n$ which converges by the Root Test but for which the Ratio Test gives no information.